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Facultad de Ingeniería

Ingeniería en Informática y Sistemas

Redes y telecomunicaciones - sección 1

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REDES Y TELECOMUNICACIONES:

LABORATORIO 9

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Paso 4 Configuración BGP en ITA

```
Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
B       10.1.1.0/24 [20/0] via 10.0.0.1, 00:00:00
    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/30 is directly connected, Serial0/0/1
B       172.16.1.0/24 [20/0] via 172.16.0.1, 00:00:00
C       192.168.0.0/24 is directly connected, Loopback0
C       192.168.1.0/24 is directly connected, Loopback1

ITA#
```

Paso 5 verificar las rutas de BGP

```
ITA#show ip bgp
BGP table version is 8, local router ID is 192.168.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop        Metric LocPrf Weight Path
*> 10.1.1.0/24      10.0.0.1              0         0   0 200 i
*> 172.16.1.0/24    172.16.0.1            0         0   0 300 i
*> 192.168.0.0/24   0.0.0.0               0         0 32768 i
*> 192.168.1.0/24   0.0.0.0               0         0 32768 i

ITA#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Cual es el local router ID

El ID del router es 192.168.1.1

Cual es la versión de la tabla que se esta desplegando

La versión de la tabla es 8

ISP1 shutdown Loopback0

```
ITA#show ip bgp
BGP table version is 9, local router ID is 192.168.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop           Metric LocPrf Weight Path
*> 172.16.1.0/24     172.16.0.1             0      0   0 300 i
*> 192.168.0.0/24    0.0.0.0                0      0 32768 i
*> 192.168.1.0/24    0.0.0.0                0      0 32768 i

ITA#
```

Cual es la versión de la table mostrada

La versión es la versión 9 debido a que se actualizo al ponerlo en shutdown uno de los lados

Que paso con la route network 10.1.1.0/24

Se desaparece de la tabla debido a que la apagamos

5e

```
ITA#show ip bgp nei
ITA#show ip bgp neighbors
BGP neighbor is 10.0.0.1, remote AS 200, external link
BGP version 4, remote router ID 10.0.0.1
BGP state = Idle, up for 00:02:39
Last read 00:02:39, last write 00:02:39, hold time is 180, keepalive interval is 60 seconds
Neighbor capabilities:
  Route refresh: advertised and received(new)
  Address family IPv4 Unicast: advertised and received
Message statistics:
  InQ depth is 0
  OutQ depth is 0

           Sent          Rcvd
Opens:          3          3
Notifications:  3          2
Updates:        8          2
Keepalives:    61         61
Route Refresh:  0          0
Total:         75         68
Default minimum time between advertisements runs is 30 seconds

For address family: IPv4 Unicast
BGP table version 10, neighbor version 6/0
--More-- %BGP-5-ADJCHANGE: neighbor 10.0.0.1 Up
Output queue size : 0
Index 1, Offset 0, Mask 0x2
1 update-group member
```

Cual es el estado de BGP

Su estado es establecido

Que tanto tiempo a estado la conexión

A estado por 00:02:39

Paso 6

6a verificación de ISP2

```
ISP2>enable
ISP2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 1 subnets
B       10.1.1.0 [20/0] via 172.16.0.2, 00:00:00
    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/30 is directly connected, Serial0/0/1
C       172.16.1.0/24 is directly connected, Loopback0
B       192.168.0.0/24 [20/0] via 172.16.0.2, 00:00:00
B       192.168.1.0/24 [20/0] via 172.16.0.2, 00:00:00

ISP2#
```

Paso 7

7a en la imagen de arriba se puede observar que no hay Gateway por eso no hay captura

7b se configuran las ip route y 7c se pone la nueva Gateway

```
ITA(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.1 210
ITA(config)#ip route 0.0.0.0 0.0.0.0 172.16.0.1 220
ITA(config)#exit
ITA#
%SYS-5-CONFIG_I: Configured from console by console

ITA#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.0.0.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
B       10.1.1.0/24 [20/0] via 10.0.0.1, 00:00:00
    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/30 is directly connected, Serial0/0/1
B       172.16.1.0/24 [20/0] via 172.16.0.1, 00:00:00
C       192.168.0.0/24 is directly connected, Loopback0
C       192.168.1.0/24 is directly connected, Loopback1
S*     0.0.0.0/0 [210/0] via 10.0.0.1

ITA#
```

7e ping de TA a ISP1

```
ITA#%BGP-5-ADJCHANGE: neighbor 10.0.0.1 Up

ITA# ping
Protocol [ip]:
Target IP address: 192.168.100.1
Repeat count [5]:
Datagram size [100]:
Timeout in seconds [2]:
Extended commands [n]: y
Source address or interface: 192.168.1.1
Type of service [0]:
Set DF bit in IP header? [no]:
Validate reply data? [no]:
Data pattern [0xABCD]:
Loose, Strict, Record, Timestamp, Verbose[none]:
Sweep range of sizes [n]:
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.100.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/7/15 ms
```

Paso 8

8b añadiendo la network 192.168.100.0/24

```
Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
C       10.1.1.0/24 is directly connected, Loopback0
       172.16.0.0/24 is subnetted, 1 subnets
B       172.16.1.0 [20/0] via 10.0.0.2, 00:00:00
B       192.168.0.0/24 [20/0] via 10.0.0.2, 00:00:00
B       192.168.1.0/24 [20/0] via 10.0.0.2, 00:00:00
C       192.168.100.0/24 is directly connected, Loopback100

ISPI#
```

8d Estableciendo la ip default-network

```
      P - periodic downloaded static route

Gateway of last resort is 10.0.0.1 to network 192.168.100.0

* 10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   10.0.0.0/30 is directly connected, Serial0/0/0
B   10.1.1.0/24 [20/0] via 10.0.0.1, 00:00:00
    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C   172.16.0.0/30 is directly connected, Serial0/0/1
B   172.16.1.0/24 [20/0] via 172.16.0.1, 00:00:00
C   192.168.0.0/24 is directly connected, Loopback0
C   192.168.1.0/24 is directly connected, Loopback1
B*  192.168.100.0/24 [20/0] via 10.0.0.1, 00:00:00

ITA#
```

8f añadir una backup route on ITA to ISP2 y conexión BGP

```
ITA#wr
Building configuration...
[OK]
ITA#
ITA#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.0.0.1 to network 192.168.100.0

* 10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   10.0.0.0/30 is directly connected, Serial0/0/0
B   10.1.1.0/24 [20/0] via 10.0.0.1, 00:00:00
    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C   172.16.0.0/30 is directly connected, Serial0/0/1
B   172.16.1.0/24 [20/0] via 172.16.0.1, 00:00:00
C   192.168.0.0/24 is directly connected, Loopback0
C   192.168.1.0/24 is directly connected, Loopback1
B*  192.168.100.0/24 [20/0] via 10.0.0.1, 00:00:00
S*  0.0.0.0/0 [220/0] via 172.16.0.1
```

8h en ves de tcslsh va a ser ping para demostrar porque no funciona el tcslsh

Ping a 10.0.0.0/24

```
ITA#ping 10.0.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/7/9 ms

ITA#ping 10.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 9/23/29 ms

ITA#ping 10.1.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/10/15 ms

ITA#
```

Ping 172.16.0.0/24

```
ITA#ping 172.16.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.0.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/10/15 ms

ITA#ping 172.16.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.0.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/21/24 ms

ITA#ping 172.16.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/14/19 ms

ITA#
```

Ping 192.16.0.0/24

```
ITA#ping 192.168.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.0.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/4 ms

ITA#ping 192.168.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/3/6 ms

ITA#ping 192.168.100.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.100.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/11/18 ms

ITA#
```

Debería ISP1 y ISP2 ser capaces de hacer ping a todas las networks en la topología

En este caso como en el paso 6 hicimos un enrutamiento con un filtro en la configuración entonces no deberían de poder hacerlo debido a que ITA no debe permitir que el trafico de ISP1 pase a través de los enlaces para llegar a ISP2 y viceversa entonces las rutas ISP deberían de poder hacer ping a ITA y a sus propias redes, pero no a las de las otras ISP